Lesson 4 - Introduction to Graphs

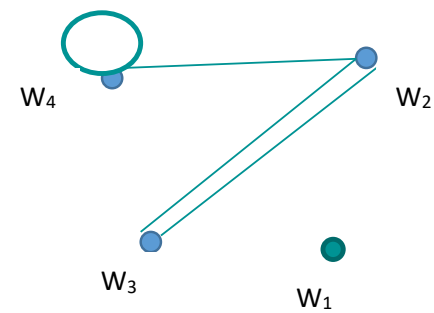
4.1 Undirected graph

Question 41

Level of difficulty: E

Given the graph (right), decide whether the following statements are true or false:

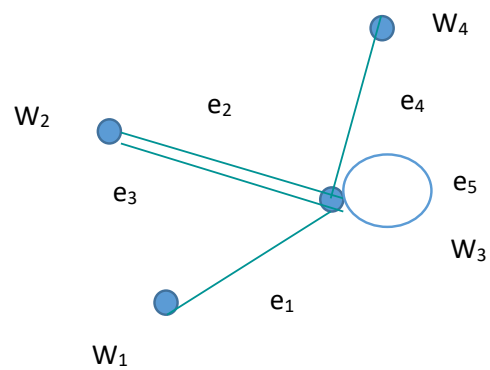
a) The graph is a simple graph.
b) The graph consists of four nodes and four edges.
c) w_3 and w_4 are adjacent.



Question 42

Level of difficulty: E

For the graph (right), give the set of nodes V, the edge set E, and the incidence mapping ϕ.



Solution question 41

Solution hint: Familiarize yourself with the definitions of graph theory.

Short solution: a) false, b) true, c) false

Detailed solution: a) False, because the graph has multiple edges and a loop, b) Correct, c) False, because both nodes are not connected by an edge.

Solution question 42

Solution hint: Familiarize yourself with the correct notation.

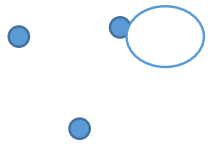
Short solution: $V = \{w_1, w_2, w_3, w_4\}$, $E = \{e_1, e_2, e_3, e_4, e_5\}$, ϕ: $e_1 \to \{w_1, w_3\}$, $e_2 \to \{w_2, w_3\}$, $e_3 \to \{w_2, w_3\}$, $e_4 \to \{w_3, w_4\}$, $e_5 \to \{w_3\}$

Detailed solution: Node set $V = \{w_1, w_2, w_3, w_4\}$, edge set $E = \{e_1, e_2, e_3, e_4, e_5\}$, incidence mapping ϕ: $e_1 \to \{w_1, w_3\}$, $e_2 \to \{w_2, w_3\}$, $e_3 \to \{w_2, w_3\}$, $e_4 \to \{w_3, w_4\}$, $e_5 \to \{w_3\}$

Question 43

Level of difficulty: M

Given the graph (right), decide whether the following statements are true or false.

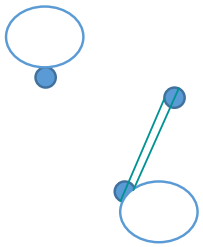


a) The graph is a finite graph.
b) The graph is a simple graph.
c) The graph is a connected graph.

Question 44

Level of difficulty: M

Given the graph (right), decide whether the following statements are true or false.



a) All nodes are adjacent.
b) The graph has a loop.
c) The graph consists of two subgraphs.

Solution question 43

Solution hint: Familiarize yourself with the definitions of graph theory.

Short solution: a) true b) true c) false

Detailed solution: a) True, since there are a finite number of nodes and edges, b) False, since it has a loop, c) False, since isolated nodes exist.

Solution question 44

Solution hint: Familiarize yourself with the definitions of graph theory.

Short solution: a) false b) true c) true

Detailed solution: a) False, because there are nodes that are not connected by an edge, b) True, because the graph has two loops, c) True, since two graphs are not connected.

Question 45

Level of difficulty: D

Given the graph (right), decide whether the following statements are true or false.

a) The graph is a simple graph.
b) It is not a graph.
c) The graph is contiguous.

Solution question 45

Solution hint: Familiarize yourself with the definitions of graph theory.

Short solution: a) true b) false c) true

Detailed solution: a) True, since there are no multiple edges and loops, b) False, since it has one node, c) True, since the graph is not isolated from another graph.

4.2 Further properties of graphs

Question 46

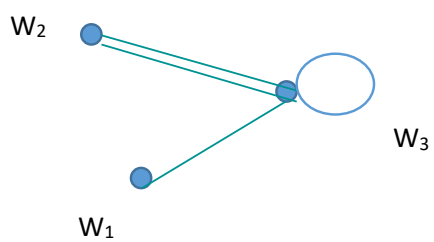
Level of difficulty: E

Draw a graph with four nodes, where one node has degree 1 and three nodes have degree 3.

Question 47

Level of difficulty: E

Determine the degree of each node.

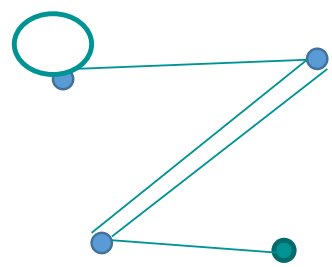


Solution question 46

Solution Note: Use multiple edges and loops if necessary.

Solution: Several solutions are possible

(loop is counted twice).



Solution question 47

Solution hint: The loop is counted twice.

Short solution: Degree of w_1: 1, degree of w_2: 2, degree of w_3: 5

Detailed solution: Degree of w_1: 1, degree of w_2: 2, degree of w_3: 5 (loop is counted twice)

Question 48

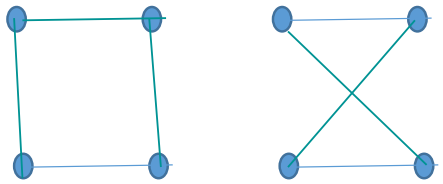
Level of difficulty: M

What do we mean when two graphs are isomorphic? Is it sufficient if they have the same number of nodes and edges?

Question 49

Level of difficulty: M

Are both graphs isomorphic? If yes, give reasons.



Solution question 48

Solution hint: Familiarize yourself with the definition of isomorphism.

Short solution: Two graphs are isomorphic if one graph is formed by redrawing the other graph; no.

Detailed solution: Two graphs are isomorphic if one is formed by redrawing the other. The edges may be bent, stretched, or contracted, but not cut or knotted. For isomorphism, it is not sufficient that two graphs have the same number of nodes, edges, and degrees.

Solution question 49

Solution hint: Familiarize yourself with the definition of isomorphism.

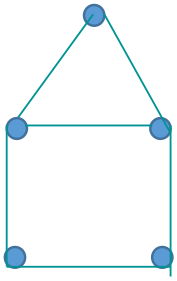
Short solution: Yes; label the edges and nodes

Detailed solution: The graphs are isomorphic. One possibility is labeling the nodes and edges so that isomorphism can be seen.

Question 50

Level of difficulty: D

Verify Euler's formula for the following graph.



Answer question 50

Solution hint: The area outside the graph is counted.

Short solution: v – e + f = 2; 5 – 6 + 3 = 2

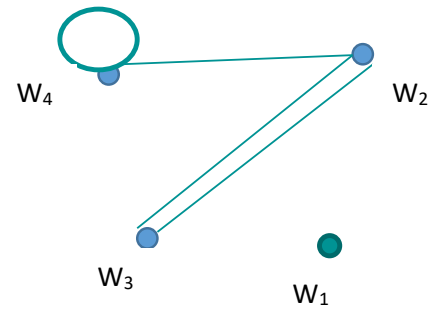
Detailed solution: Number of nodes v – number of edges e + number of faces f = 5 – 6 + 3 = 2. The result corresponds to Euler's formula.

4.3 Adjacency matrix

Question 51

Level of difficulty: E

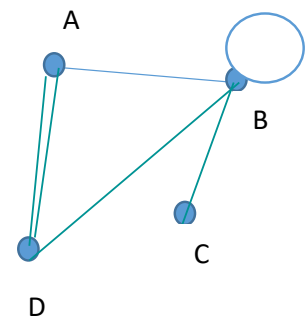
Given the following graph, state the adjacency matrix.



Question 52

Level of difficulty: E

Given the following graph, state the adjacency matrix.



Solution question 51

Solution hint: Loops are counted twice.

Solution: A = $\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 1 \\ 0 & 2 & 0 & 0 \\ 0 & 1 & 0 & 2 \end{pmatrix}$

Solution question 52

Solution hint: Loops are counted twice.

Solution: A = $\begin{pmatrix} 0 & 1 & 0 & 2 \\ 1 & 2 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 2 & 1 & 0 & 0 \end{pmatrix}$

Question 53

Level of difficulty: M

A graph is given by the following adjacency matrix. Make a drawing.

$$A = \begin{pmatrix} 2 & 1 & 2 & 1 \\ 1 & 0 & 0 & 0 \\ 2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}$$

Question 54

Level of difficulty: M

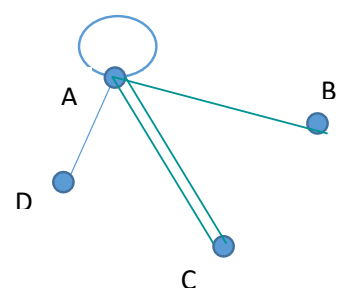
A graph is given by the following adjacency matrix. Make a drawing.

$$A = \begin{pmatrix} 2 & 1 & 2 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 2 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Solution question 53

Solution hint: First draw the nodes A, B, C and D.

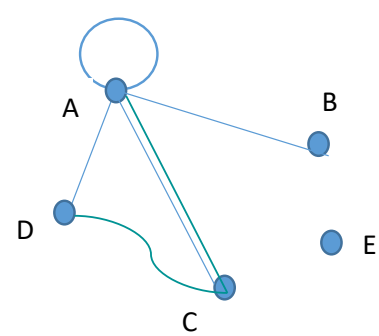
Solution: See graph



Solution question 54

Solution hint: First draw the nodes A, B, C, D and E.

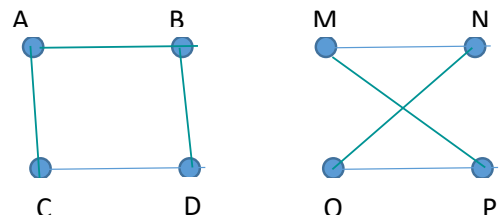
Solution: See graph



Question 55

Level of difficulty: D

Check against the adjacency matrix, whether the graphs are isomorphic to each other.



Solution question 55

Solution hint: To do this, give a bijective mapping of the set of nodes of G to the set of nodes of H and check that the two adjacency matrices are equal.

Solution: The two graphs are isomorphic. A -> M, B -> N, C -> P, D -> O

$A_{left} = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix}$ $A_{right} = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix}$